Reconfigurable Model Predictive Control for Articulated Vehicle Stability With Experimental Validation

Yubiao Zhang, Amir Khajepour, Ehsan Hashemi, Yechen Qin, *Member, IEEE*, and Yanjun Huang

***Abstract*—This article proposed a reconfigurable control scheme for articulated vehicles' stabilization by leveraging optimization-based control techniques. The central objective is to maintain a good lateral and yaw stability of the vehicle with optimal corrective brakes, meanwhile applicable to different actuation configurations. This is achieved by a two-layer control structure, where the high-level controller formulates as a model predictive control (MPC) tracking problem to generate corrective center-of-gravity (CG) yaw moment of each unit. The lower level controller utilizes the control allocation (CA) algorithm with real-time constraints to optimally calculate differential brakes at each wheel with maximum utilization-of-tires capacity. To evaluate its real-time performance, experimental validation is carried out on the electrified tractor-trailer with selective differential braking systems. It is observed that the controller is effective in dynamics control, meanwhile reconfigurable to various actuation configurations. Furthermore, the proposed system has great potential in production tractor-trailer systems due to the low cost and number of sensor requisites.**

***Index Terms*—Control allocation (CA), electrified articulated vehicles, model predictive control (MPC), real-time experiment, reconfigurable control, transportation safety.**

I. Introduction

ARTICULATED vehicles, such as trucks and car trailers, have been playing a vital role in trucking transportation and family market. In North America, car-trailer combinations, usually an SUV or pickup truck towed with a trailer, are widely used to transport goods and materials due to their low cost and high versatility [1]. However, articulated vehicles differ from the passenger car quite a lot, and it makes a big challenge for dynamics controls. Characterized by a high wheelbase to track ratio, a high center of gravity (CG), and uneven or changeable weight distributions, articulated vehicles are prone to instability, such as trailer sway and jackknife or even rollover [2], [3]. There are more than 5000 deaths in truck accidents in 2017 alone [4] and over 20 000 people injured every year in accidents involving utility trailers towed by passenger vehicles [5]. To enhance transportation safety, a great deal of theoretical and experimental work is done for vehicle dynamics control or driver-assistance systems [6]–[9]. Implemented solutions, such as antilock braking system (ABS) and electronic stability program (ESP), have saved many lives today. Hac *et al.* [10] used active trailer braking to control dynamics and stability of an articulated vehicle in the yaw plane. Lee *et al.* [11] proposed an optimal robust controller for car trailers using active trailer differential braking. Chen and Shieh [12] investigated the use of differential brake to prevent jackknifing of tractor-semitrailer using the model reference adaptive control. In a more advanced way, Kayacan *et al.* [13] used the robust model predictive control techniques to address the trajectory tracking problem of an autonomous tractor-trailer. Despite this, active safety systems are mainly for passenger cars, and these for articulated vehicles and electric tractor-trailer are scarce.

Vehicles are usually overactuated mechanical systems, e.g., four- ora more than four-wheel providing drive/brake torques. In addition, tractor-trailer may have various active control actuation systems depending on needs and feasibility, for instance, active steering and differential torques (drive or brake). An ideal controller should be reconfigurable and selective to different configurations and make the best use of them. To deal with it, integrated chassis control [14]–[16], reconfigurable control, and control allocation (CA) have gained much attention [17]–[19]. Integrated control makes the most use of all actuation systems and prevents conflicts of different objectives and subsystems. Reconfigurable control provides much freedom for controller design to fit different configurations of vehicle so that significant modification and major tuning can be avoided. Moreover, the advances in control theory and autohardware, e.g., computing systems, make it possible to implement complex control systems in real time. For example, model predictive control (MPC) for vehicle control and applications has attracted extensive attention [20], [21]. MPC has the unique capability of considering the inputs and output constraints explicitly and providing optimal (locally) solution in real time. Taken in the context of the vehicle stability control problem, the objective of dynamics control is to track some desired states or reduce to certain threshold values. The outputs are usually bounded for safety concerns, and actuators have physical limitations or tires capacity. Therefore, the problem setting is well-matched to the

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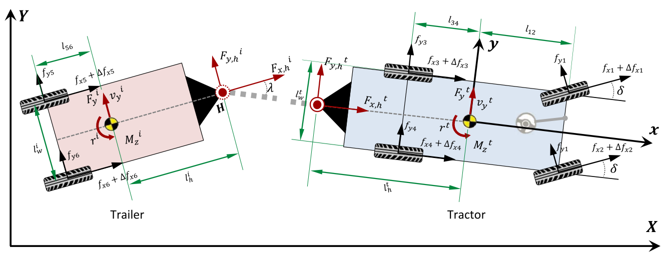

Fig. 1. Separated units of an articulated vehicle.

MPC formulation. Some representatives of MPC to vehicle stability control applications can be found from many studies [16], [18], [22], [23]. As another technique, CA offers the advantage of overactuated systems to allocate the desired generalized controls (typically virtual) among all active actuators optimally. Important issues, such as input saturation and rate constraints, actuator and fault tolerance, and minimal control energy, are handled within the CA algorithm [24].

As a core contribution, this article develops a reconfigurable model predictive controller for articulated vehicle stability improvement and validates with different actuation systems in real-time implementation. The controller is able to address issues of multiple control objectives, control energy minimization and actuation diversity, and redundancy of articulate vehicles. Moreover, considering that most trailers are passive systems, the promising performance of tractor differential braking strategy gives us much evidence and confidence to solely use the tractor actuators to control both units, yet only using the articulation angle sensor of the trailer. Experimental validation is conducted on an articulated vehicle with multiple configurations.

In the remainder of this article, Section II describes the articulated vehicle modeling using an intuitive manner. Section III formulates the high-level MPC and lower level CA controller and discusses real-time constraints and reconfigurability. Section IV gives real-time implementation and experimental results. Section V concludes this article.

II. Control-Oriented Modeling

Articulated vehicles' dynamics are complex due to the coupled constraints. Many tried to model it directly through differential equations and complex transformations and items elimination. Our idea is simple and straightforward. Intuitively, the dynamics of articulated vehicles (one articulation) is "dynamics of two separate units (tractor and trailer) + constraints at articulation point" (see details in [25]). Tractor-trailer dynamics in the yaw plane are illustrated in Fig. 1. Assuming a constant longitudinal speed, the articulated vehicle has three DoFs considered in the yaw plane, where the tractor unit has the freedom of side-slip, yaw motion, while the trailer has the freedom of yaw motion relative to the tractor.

A. Detached Tractor and Trailer

Vehicles with a single unit are widely studied and utilized for stability control (see [26]) which is not repeated here. We detach the articulated vehicle as two units in Fig. 1, where each unit is an independent vehicle. The dynamics of the tractor unit for lateral and yaw motions can be described as

$$m^t\left(\dot{v}_y^t + r^t v_x^t\right) = F_y^t + F_{y,h}^t \tag{1a}$$
$$I_{zz}^t \dot{r}^t = M_z^t - l_h^t F_{y,h}^t + \Delta M_z^t \tag{1b}$$

where the CG lateral force F_y^t and CG yaw moment M_z^t can be mapped form tire forces

$$\begin{aligned} F_y{}^t &= f_{x1}\sin\delta + f_{y1}\cos\delta + f_{x2}\sin\delta \\ &\quad + f_{y2}\cos\delta + f_{y3} + f_{y4} \end{aligned} \tag{2a}$$
$$\begin{aligned} M_z{}^t &= l_{12}(f_{x1}\sin\delta + f_{y1}\cos\delta + f_{x2}\sin\delta + f_{y2}\cos\delta) \\ &\quad - l_{34}(f_{y3} + f_{y4}) + 0.5l_w^t(f_{x2}\cos\delta \\ &\quad + f_{x4} - f_{x1}\cos\delta - f_{x3}). \end{aligned} \tag{2b}$$

Similarly, the dynamics equations of the trailer unit are written as

$$m^i\left(\dot{v}_y^i + r^i v_x^i\right) = F_y^i + F_{y,h}^i \tag{3a}$$
$$I_{zz}^i \dot{r}^i = M_z^i + l_h^i F_{y,h}^i + \Delta M_z^i \tag{3b}$$

where the CG lateral force F_y^i and CG yaw moment M_z^i can be mapped form tire forces

$$F_y{}^i = f_{y5} + f_{y6} \tag{4a}$$
$$M_z{}^i = -l_{56}(f_{y5} + f_{y6}) + 0.5l_w^i(f_{x6} - f_{x5}). \tag{4b}$$

From (1)–(4), parameters/variables with superscript t denote the tractor and i the trailer. Vehicle's parameters are described in Section IV. The vehicle states are the wheels' steering angle δ, the longitudinal velocity v_x^t/v_x^i, the lateral velocity v_y^t/v_y^i, the yaw rate r^t/r^i, and the force at articulation point $F_{y,h}^t/F_{y,h}^i$ while decoupling two units. $\Delta M_z^t/\Delta M_z^t$ is the corrective CG yaw moment; $f_{x1}, f_{x2}, \ldots, f_{x6}$ and $f_{y1}, f_{y2}, \ldots, f_{y6}$ are the tires longitudinal and lateral forces from 1 to 6, as shown in Fig. 1.

B. Articulation Constraints

Two groups of constraints are suggested at the articulation point. First, the kinematic constraint, namely, the velocities at the CG of the trailer, can be written with respect to the velocities at the CG of the tractor

$$v_x^i = v_x^t\cos\lambda + \left(v^t - l_h^t r^t\right)\sin\lambda \tag{5a}$$
$$v_y^i = -v_x^t\sin\lambda + \left(v_y^t - l_h^t r^t\right)\cos\lambda - l_h^i r^i \tag{5b}$$

where λ is the articulation angle (art. angle, for short), and it yields $\dot{\lambda} = r^i - r^t$. Second, the dynamics constraints, namely, the relationship of the longitudinal and lateral force between the tractor and trailer at the articulation point, are written as

$$F_{x,h}^t = -F_{x,h}^i\cos\lambda - F_{y,h}^i\sin\lambda \tag{6a}$$
$$F_{y,h}^t = F_{x,h}^i\sin\lambda - F_{y,h}^i\cos\lambda. \tag{6b}$$

C. State Space Formulation

A brush tire model is used to calculate the tire lateral force. To reduce the computation load, yet reserve model fidelity, the brush tire model is linearized at the slip angle operating point by taking the differentiation, and an affine model is formulated

$$f_{yi} = \bar{f}_{yi} - \bar{C}_{\alpha i}(\alpha_i - \bar{\alpha}_i), \quad (i = 1, \cdots, 6) \tag{7}$$

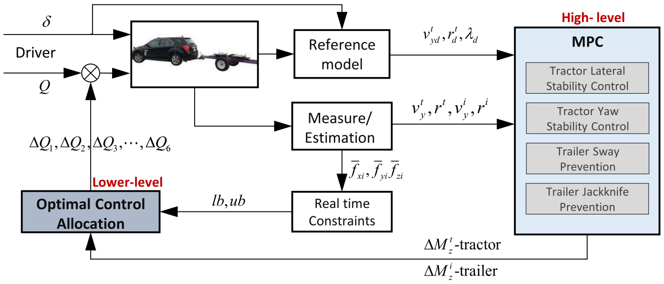

Fig. 2. Structure of the reconfigurable controller.

where, at the operating point ($\bar{\alpha}_i$), the tire force ($\bar{f}_{yi}$) and the equivalent cornering stiffness ($\bar{C}_{\alpha i}$) can be found through the brush model formed in lookup tables. The linearizing technique connects the vehicle state and tire force, which results in a linear-time-varying (LTV) model and allows the MPC controller to formulate the predictions near the operating point as well as considering tire saturation area. By the restructuring the algebra equations (1)–(7) and only retaining the vehicle states for control synthesis, the vehicle dynamics can be described by the following compact state-space formulation, where details can be found in [27]:

$$\dot{x} = A_v x + B_v v + G_v \delta + d_v \tag{8a}$$
$$y = C_v x \tag{8b}$$

where the vehicle state vector is $x = [v_y^t \ r^t \ v_y^i \ r^i]^T$, the high-level control input vector is $v = [\Delta M_z^t \ \Delta M_z^i]^T$, and the output vector is $y = [v_y^t \ r^t \ \lambda]^T$.

III. Reconfigurable Controller Design

The overall control structure with two layers is illustrated in Fig. 2. The driver's steering and drive/brake torque commands are passed to the vehicle as a feedforward input; meanwhile, go to the reference model module to generate the desired states. To have a dynamic model prediction, the sensing and estimation module provides reliable state variables to the high-level MPC for model prediction. The MPC formulation minimizes the errors between the reference sequences and sequences predicted in the assumed control horizon using current state and driver inputs. Required CG corrections, i.e., yaw moment $\Delta M_z{}^t$ and $\Delta M_z{}^i$, are computed online by solving a quadratic optimization problem. Now, feeding them into the lower level CA, the augmented torque or braking ($\Delta Q_1, \Delta Q_2, \cdots, \Delta Q_6$) is optimally distributed by solving a constrained quadratic problem at each time step.

Remark: Although the differential braking strategy is considered in this article, the control scheme is able to work with active steering or torque vectoring or any integrated control. To explain why to use two layers, vehicles have various configurations, but the corrective CG forces are universal. This virtual control effort does not need prior knowledge about the actuators. In such a way, the lower layer controller handles the active actuators' redundancy and diversity. Furthermore, in an MPC control algorithm, the computational cost grows significantly if using brake actions, which may have six variables in this application. It is a huge burden in real time at automotive-grade hardware.

A. High-Level MPC

The high-level MPC is designed to compute the corrective CG moments for stability's sake. MPC is naturally capable of handling multivariable systems, which simplifies the controller development when multivariables and different control objectives need to be considered. Furthermore, states and actuator constraints are easily formulated into the optimization problem. MPC has inherent local robustness to disturbances and uncertainties [20]. Desired states of the lateral speed, the yaw rate, and the articulation angle are generated and updated from the reference module

$$y_d = \begin{bmatrix} v_{y_d}^{\ t} & r_d^t & \lambda_d \end{bmatrix}^T \tag{9}$$

where the desired sates are defined from vehicle stable and steady-state behaviors seen in [19], and [26].

The LTV model is discretized using the Euler method (zero-order holder) [28]

$$x(k+1) = A_d x(k) + B_d v(k) + d_w(k) \tag{10a}$$
$$y(k) = C_d x(k) \tag{10b}$$

where $d_w(k) = G_d \delta(k) + d(k)$ for computation's sake. The cost function is defined in a finite horizon N_p and a control horizon N_c

$$J(x_{0,t}, V_t) = \sum_{k=1}^{N_p} \|y_{t+k,t} - y_{d_{t+k,t}}\|_Q^2 + \sum_{k=0}^{N_c-1} \|v_{t+k,t}\|_R^2 + \|v_{t+k,t} - v_{t+k-1,t}\|_T^2 \tag{11}$$

where $V_t = [v_{t,t}, \ \cdots \ v_{t+N_c-1,t}]^T, \in \mathbb{R}^{2N_c \times 1}$, represents the optimization sequence at time t. Index $t+k,t$ denotes the predicted value at k steps ahead of the current time t. The first term in (11) is the tracking error of the references. The second term is the prospective CG forces and minimizes the control efforts. The third term is optional and enforces proximity to the previous step to prevent oscillation in control actions. The positive semidefinites Q, R, and T are the weighting matrices that reflect the importance of these terms in the cost function. At each time step, the following finite horizon (N_p) optimal control problem is solved online:

$$\begin{aligned} \min_{V_t} \ & J(x_{0,t}, V_t) \\ \text{s.t.} \ & x_{k+1,t} = A_d x_{k,t} + B_d v_{k,t} + d_{wk,t} \\ & y_{k,t} = C_d x_{k,t} \\ & y_{\min} \le y_{k,t} \le y_{\max} \\ & k = 1, \cdots, t+N_p \\ & v_{\min} \le v_{k,t} \le v_{\max} \\ & k = 0, \cdots, t+N_c-1. \end{aligned} \tag{12}$$

Given the expansion of the prediction model [27], a Batch approach in [29] is used to formulate the problem as a standard QP problem. The objective function can be expressed as a function of the initial state $x_{0,t}$ and sequences of reference, driver input. The solver, i.e., *qpOASES*, based on a structure-exploiting active-set method, provides fast and reliable solutions and is used in this article [30]. The feasibility and stability of the LTV-MPC scheme are well proved and discussed in [21]. The corrective control vector should be bounded by the maximum CG moment that a specific vehicle can generate. The high-level controller, hence, completes the calculation of optimal corrective GG moments.

B. Lower Level Optimal CA

In this section, calculations from high-level MPC are applied through differential braking. Since articulated vehicles are overactuated mechanical systems using multiple braking wheels, a CA is used to cope with such redundant actuation systems and various configuration of active actuators [17], [24]. As a result, CA offers an optimal distribution on actuator commands, meanwhile the minimization of the CG corrective forces errors.

The desired CG yaw moments are obtained by interpreting the driver's desires and are denoted by

$$M_z^* = \begin{bmatrix} M_z^{t*} & M_z^{i*} \end{bmatrix}^T \tag{13}$$

where M_z^{t*} and M_z^{i*} are the desired CG yaw moments of tractor and trailer, respectively. The actual CG yaw moment acting on the tractor-trailer is written as

$$M_z = \begin{bmatrix} M_z^{t} & M_z^{i} \end{bmatrix}^T \tag{14}$$

where M_z^t and M_z^i are the actual CG yaw moments of tractor and trailer, respectively. These states cannot be measured/estimated but serve to demonstrate the development of the corrective yaw moment at CG. Since this article focuses on differential braking systems and the lateral tire forces are not actively controlled, we define the tire force vector that only including the longitudinal forces

$$f = [f_{x1} \quad f_{x2} \quad \cdots \quad f_{x6}]^T, \quad f \in \mathbb{R}^{6\times 1}. \tag{15}$$

Each CG yaw moment is a function of all tire forces, namely, $M_z^t(f)$ and $M_z^i(f)$. In this case, the linear transformation represented by $\mathbf{J_{M_z}}$ gives the best linear approximation of M_z near the point f

$$M_z(f + \Delta f) \approx M_z(f) + \mathbf{J_{M_z}} \Delta f \tag{16}$$

where $\mathbf{J_{M_z}}$ is the Jacobian Matrix of M_z that maps the tire forces to CG yaw moment. Therefore, the corrective yaw moments are mapped by $\mathbf{J_{M_z}} \Delta f$. Only considering tire longitudinal forces, $\mathbf{J_{M_z}}$ is obtained by

$$\mathbf{J_{M_z}} = \begin{bmatrix} \dfrac{\partial M_z^t}{\partial f_{x1}} & \dfrac{\partial M_z^t}{\partial f_{x2}} \cdots & \dfrac{\partial M_z^t}{\partial f_{x6}} \\ \dfrac{\partial M_z^i}{\partial f_{x1}} & \dfrac{\partial M_z^i}{\partial f_{x2}} \cdots & \dfrac{\partial M_z^i}{\partial f_{x6}} \end{bmatrix}. \tag{17}$$

The corrective CG yaw moments are defined as the difference between (13) and (14), and they are also calculated from the high-level MPC

$$\Delta M_z^* = M_z^* - M_z. \tag{18}$$

Therefore, the lower level CA problem is formulated as a convex quadratic programming with linear constraints. At each time step, it solves the following optimization problem:

$$\begin{aligned} &\min_{\Delta f} \ \xi \| v^* - \mathbf{J_{M_z}} \Delta f \|_{W_e}^2 + \| \Delta f \|_{W_{\Delta f}}^2 \\ &\text{s.t. } \mathrm{lb} \le \Delta f \le \mathrm{ub} \end{aligned} \tag{19}$$

TABLE I

FLOWCHART OF THE ITERATION ALGORITHM

High-level MPC
1. Acquire the new state $x(t)$, driver input $w(t)$, reference state $y_d(t)$;
2. Obtain V_t^* by solving the QP optimization problem (12) ;
3. Apply $v(t) = v_{0,t}^*$ (the first element of V_t^*) to the lower-level controller;
Lower-Level Optimal CA
4. Acquire the $v(t) = v_{0,t}^*$;
5. Obtain Δf^* by solving the static QP optimization problem (19) ;
6. Apply ΔQ^* to the brake actuators using Δf^* ;
7. $t \leftarrow t + 1$. Go to 1.

where the first term in the objective function of (19) is to minimize the CG yaw moments error, while the second term is to minimize the control efforts of actuators. lb, ub are the lower and upper bounds vector, and $W_e, W_{\Delta f}$ are the positive definite weighting matrices, which gives a compromise between two costs. ξ is used to emphasize the importance of minimizing the allocation error that typically set very large. In addition, $W_e, W_{\Delta f}$ may be time varying to realize actuators prioritization that reflects how the available actuators will be preferred and utilized in real time, which is omitted for sake of brevity. To summarize the high-level controller and lower level controller, a flowchart demonstrates how they works in one cycle shown in Table I.

C. Constraints and Reconfigurability

Given the calculation of the corrective longitudinal forces Δf, the differential brakes applied to each wheel are calculated as follows considering small slip ratios:

$$\Delta Q_i = \begin{cases} R_{\text{eff}}^t \Delta f_i, & i = 1, \cdots, 4 \\ R_{\text{eff}}^i \Delta f_i, & i = 5, 6 \end{cases} \tag{20}$$

where R_{eff}^t and R_{eff}^i are wheel effective radius of the tractor/trailer and brakes vector $u = [\Delta Q_1, \Delta Q_2, \cdots, \Delta Q_6]^T$. In this application, lb(i)/ub(i) denotes the lower/upper boundary of diff. brake force of wheel i

$$\mathrm{lb}(i) = \max\left(f_{xi}^{\text{brake}}, -\bar{f}_{xi}^{\text{tire}}, -\bar{f}_{xi}^{\text{fail}}\right) - \bar{f}_{xi} \tag{21a}$$

$$\mathrm{ub}(i) = \min\left(-\bar{f}_{xi}^{\text{tire}}, -\bar{f}_{xi}^{\text{fail}}\right) - \bar{f}_{xi} \tag{21b}$$

where $\bar{f}_{xi}$ is the estimated force of wheel i from the driver's request torque and f_{xi}^{brake} is the maximum brake torque depending on the vehicle's braking system. $\bar{f}_{xi}^{\text{tire}}$ is derived from a "tire ellipse model" [31]. Given the estimated tire forces, i.e., $\bar{f}_{xi}$, $\bar{f}_{yi}$, and $\bar{f}_{zi}$, the maximum "available longitudinal force" is achieved. Although wheel slip prevention is not included in this article, it can be tackled by off-the-shelf modules, e.g., ABS. Furthermore, the tire capacity consideration is capable of preventing wheels slips. $\bar{f}_{xi}^{\text{fail}}$ is the value when an actuator failure occurs. CA is able to handle fault-tolerant control due to updating constraints in real time. Once a fault is detected, CA will systematically set the faulty value to the bounds of the failed wheel. There is no need to redesign the control laws as it can allocate the remaining actuators automatically to best meet the control tasks.

The reconfigurability of different brake configurations is achieved by assigning the bounds and weights of the corresponding inputs. There are three differential braking strategies

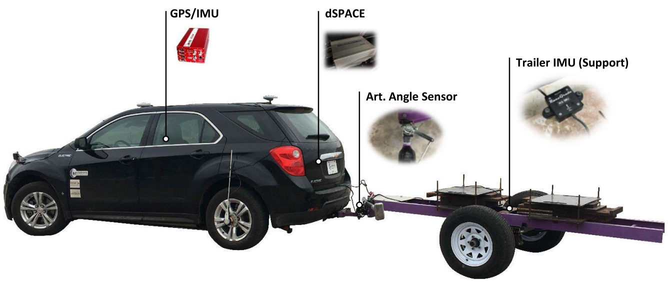

Fig. 3. Test tractor and customized trailer, and main hardware.

examined in this article. For instance, tractor differential braking only generates corrective torques ΔQ_1 to ΔQ_4, which is defined as "Controller A." To configure the controller, we assign the weights and bounds of ΔM_z^t in the MPC controller to reasonable values (see in Table III); and the weight of ΔM_z^i is set to be large enough, and bounds are very small, i.e., a slack value. For the lower level CA, the upper bounds ub(5) and ub(6) are zeros, while lower bounds lb(5) and lb(6) are negatively small. Following this mathematical modification, the controllers of trailer differential braking (defined as "Controller B") and integrated differential braking (defined as "Controller C") are also realized.

IV. Real-Time Implementation and Results

To validate the effectiveness and reconfigurability of the controller in real time, experimental studies are conducted on a tractor-trailer in this section. Three different control actuation systems are configured, tested, and compared.

A. Experimental Facilities

1) Tractor: Electrified vehicles not only provide a promising future on energy and climate issues but also have a significant impact on the chassis system and active actuation controls [32]. The test tractor is a pure electric four-wheel independent drive Chevrolet–Equinox (2011), as shown in Fig. 3, which is electrified compared with its gasoline version. It allows each wheel to be controlled independently, which applies to different driveline configurations, such as FWD, RWD, and 4WD. More importantly, the configuration provides great benefits on active torque vectoring and regenerative brake control that is used in this article. Each motor has up to ±1600 Nm, and an ABS module is available on this vehicle for wheel slip prevention. The main parameters of the tractor are listed in Table II.

2) Trailer: The trailer, as shown in Fig. 3, was designed and made for active trailer control purpose. To cope with the significant effect of configuration variation, the trailer is designed to be adjustable in payload and CG location. It can be seen from Fig. 3 that there are two removable/addable clump weights between the trailer axle and the axle, which can be fixed at different positions over the frame through bolting. Trailers #1 and #2 are configured as stable and unstable configurations by changing the payload and its distribution. The main parameters of these two trailers are listed in Table II. In addition, the trailer is equipped with electronic drum brake system that can be controlled independently. It gives freedom to test the trailer differential brake strategy.

TABLE II
Dynamics Parameters of Tractor-Trailer

Description (Symbol)	Unit	Tractor	Trailer#1	Trailer#2
Total mass (m^t, m^i)	kg	2270	650	700
Yaw moment of inertia (I_{zz}^t, I_{zz}^i)	kg.m^2	4600	900	1200
Distance of front axle to CG (l_{12})	m	1.42	N/A	N/A
Distance of rear axle to CG (l_{34}, l_{56})	m	1.44	0.21	-0.1889
Average track width (l_w^t, l_w^i)	m	1.59	1.32	1.32
Distance of hitch point to CG (l_h^t, l_h^i)	m	2.54	1.89	1.889
Wheel effective radius (R_{eff}^t, R_{eff}^i)	m	0.34	0.40	0.40
Understeer coefficient (K_{us}^t, K_{us}^i)	-	0.004	—	-1.28e-4

3) Sensors/Hardware: Sensing is a critical part of providing vehicles' states and feedbacks for closed-loop control implementation. Most of the sensors are pointed and illustrated in Fig. 3. The steering wheel angle sensor, located in the steering column, is to provide steering angle and rate of turn. The global positioning system (GPS) is installed on the tractor to measure the longitudinal speed and lateral speed of the vehicle accurately. A six-axis inertial measurement unit (IMU) is mounted inside of the tractor, close to the CG location. To acquire the information of trailer yaw motion, the articulation angle sensor is needed, and it is the only necessary sensor of the trailer in this control system. It is mounted right at the articulation point to provide articulation angle and its rate. Although not a requisite, an additional autograde IMU is mounted in the trailer as supported module to provide trailer yaw rate, lateral acceleration, and so on. All these measurements and controller calculations to the vehicle are communicating through a dSPACE MicroAutoBox II, as shown in Fig. 3. The proposed controller is developed in MATLAB/Simulink environment and then complied in dSPACE to realize the real-time implementation and control.

There are some states that cannot be used directly due to noises and corruption; therefore, estimation and filter are used instead. The articulation angle and its rate go through a median filter and then fed to the controller. The tire forces (longitudinal forces, lateral forces, and normal forces) of the tractor are estimated for constructing the tire constraints [33]. Since the trailer is usually a passive system with limited sensors, the trailer tire constraints are only bounded by brake capacity. There comes a computational challenge while implementing MPC controller. Some techniques are used to address it: 1) using the computed result of the last step as current input while the optimization is not solved within the time constraint; 2) by constraining the max iteration of the QP solver so that there is sufficient computation for the real-time optimization; and 3) using proper prediction horizon and control horizon along with the sampling time.

B. Tractor Differential Braking

The control objective is to maintain good lateral and yaw stability for the tractor and prevent the trailer from sway. The unstable configuration of Trailer #2 is used in tests as it is more likely to trigger a trailer sway at low speeds. Continuous sine steer and double-lane change, as two major maneuvers, are conducted on a half-wet pavement, and the controller is expected to be robust to road friction variation. For each case with active controller engaged, a highly similar but open-loop (Control OFF) maneuver is performed for comparison purposes.

TABLE III
PARAMETERS OF RECONFIGURABLE CONTROLLER

Symbol	Description	Value
T_s	Controller sample time (s)	0.01
N_p	Steps of the prediction horizon	10
N_c	Steps of the control horizon	5
α^{sat}	Tire saturated side slip angle (°)	7.5
q_1	Weight of lateral speed tracking error	32.4
q_2	Weight of yaw rate tracking error	900
q_3	Weight of art. angle tracking error	3600
r_1	Weight of tractor yaw moment	5e-07
r_2	Weight of trailer yaw moment	2e-06
$\Delta M^t_{z,min}$	Bounds of tractor yaw moment (Nm)	-5000
$\Delta M^t_{z,max}$	Bounds of tractor yaw moment (Nm)	5000
$\Delta M^i_{z,min}$	Bounds of trailer yaw moment (Nm)	-1500
$\Delta M^i_{z,max}$	Bounds of trailer yaw moment (Nm)	1500
W_e	Weight of CG forces mapping error	diag([50,50])
W_u	Weight of control efforts	0.02*diag([1,1,1,1,2,2])
Q^t_{brk}/Q^i_{brk}	Tractor/trailer brake limits (Nm)	-1500/-800

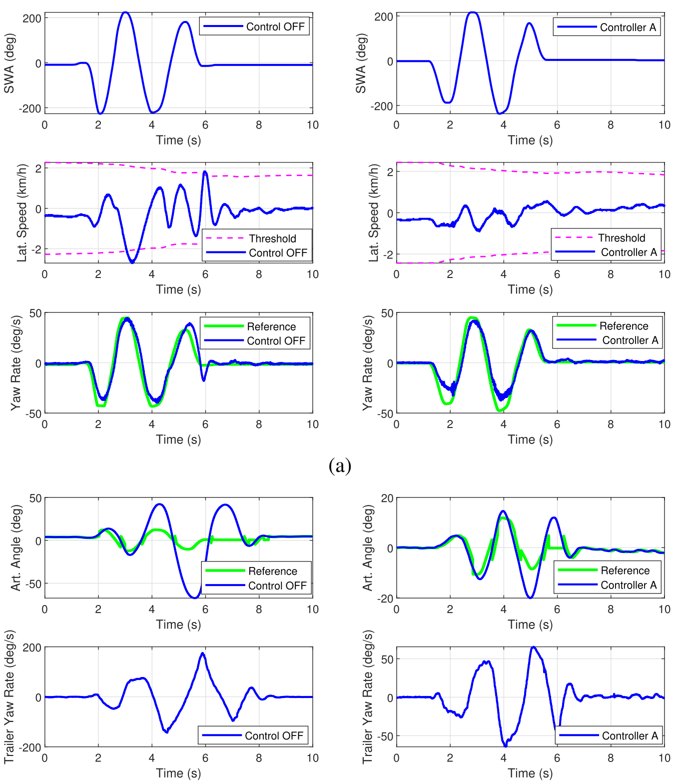

Fig. 4. Tractor-trailer responses for Control OFF and Controller A in sine steer. (a) Tractor comparative results. (b) Trailer comparative results.

We started the tests with a tractor regenerative differential braking system. To realize this strategy and only calculate the corrective torque from ΔQ_1 to ΔQ_4, one could modify the gains and bounds of the MPC and CA listed in Table III. Both sine steer and DLC maneuvers are performed with and without the controller for comparison.

Figs. 4 and 5 give the results of sine steer maneuver. In Fig. 4, it is easy to find all subfigures at the left column denote the "Control OFF" maneuver, while these at right column denote the "Controller A" engaged. At the top of the figure, it is shown that the driver performed sine steering and dropped the throttle and brake at the same speed of 48 km/h. The steering angles applied for both maneuvers are almost the same in spite of small discrepancies. Fig. 4(a) compares the lateral speed and yaw rate of the tractor, while Fig. 4(b) compares the articulation angle and yaw rate of the trailer. In the Control OFF case, the lateral speed is generally much larger than that in Controller A and even exceeds the permissible boundary at 3 and 6 s. The lateral speed with control remains very small and far less than the boundary limit. The yaw rate behaved stable and similar in both cases, and it suggested that the tractor is more understeer than the reference derived from an understeer characteristic. This is mainly due to load transfer effect of the unstable trailer. Looking at the trailer results in Fig. 4(b), while no control engaged, the art. angle is deviated from the reference harshly after 3.5 s and reached almost 40° then reached to the peak value of $-65°$ at 5.5 s. The whole sway lasted from 2 to 8 s. Trailer yaw rate shows similar evidence that the range went $-150°/s$–$190°/s$. Clearly, this is a very unstable and dangerous situation. On the contrary, in the Controller A ON case, it is shown that art. angle is significantly reduced to less than 20° and lasted till 6.5 s. The vehicle tests are filmed by a drone, and one could find the visualized and intuitive comparison from the footage in Fig. 6 that the trailer is swaying harshly without control. The trailer yaw rate also dropped to a reasonable range that is close to the tractor yaw rate. The corrective (braking) torques of the tractor are shown in Fig. 5. It can be seen that the torques are smooth, and the transitions from side to side are consistent. In addition, the torques requested are within the actuator and tire capacity due to the real-time constraints.

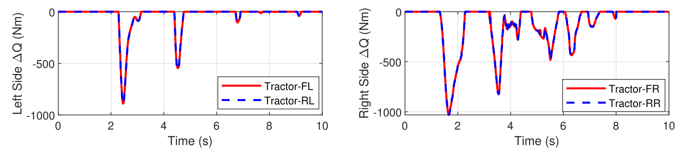

Fig. 5. Corrective torques by Controller A in sine steer.

Figs. 7 and 8 present the results of DLC maneuver. For safety concerns, the initial speed is set at 45 km/h, and the driver applied the similar steering angles of DLC. Fig. 4(a) compares the lateral speed of the tractor, while Fig. 4(b) compares the articulation angle of the trailer. In the Control OFF case, the lateral speed exceeded the bounds and reached -4km/h at 4 s, which is equivalent to $-6°$ of the CG sideslip angle. The yaw motion is stable for both cases although not shown here. The responses with controller A engaged behaved better and kept in stable margin. Again, the significant improvement is seen in trailer responses due to the requested differential torques shown in Fig. 8. Trailer art. angle is reduced from 40° to 20°, and its yaw rate is around 40% decreased; thus, the sway is stabilized successfully.

C. Trailer Differential Braking

Due to the reconfigurability of the proposed controller, we could easily move from the tractor differential braking to trailer differential braking without reformulating the problem again. To realize this strategy (Controller B) and only calculate the corrective torque of ΔQ_5 and ΔQ_6, one could modify the gains and bounds of the MPC and CA listed in Table III.

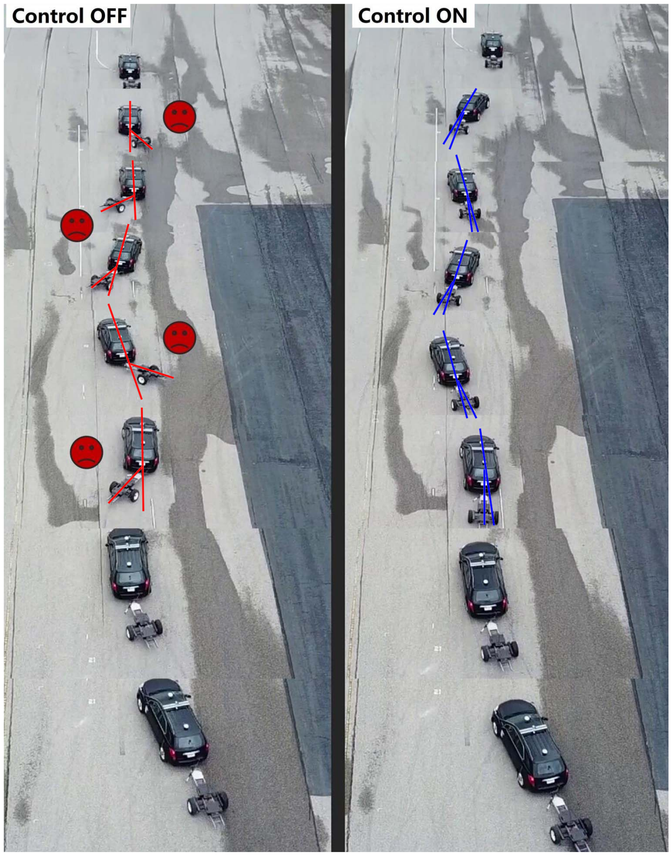

Fig. 6. Footage for Control OFF and Controller A in sine steer (the "sad" face marks the moments when the trailer is experiencing large art. angles without control; the cross lines demonstrate and compare the art. angles in selected moments of Control OFF and ON).

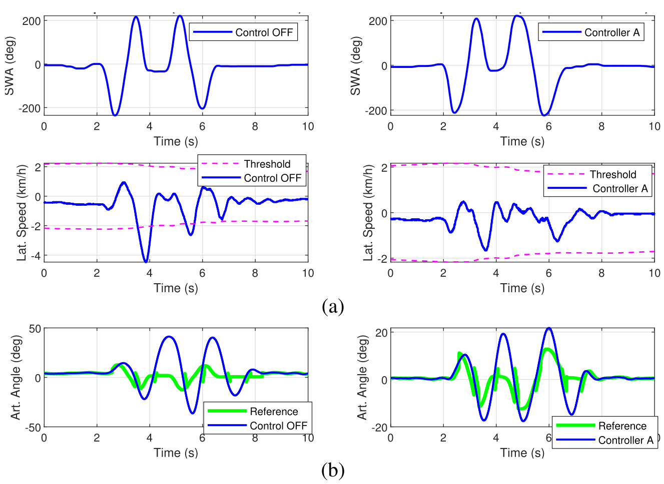

Fig. 7. Tractor-trailer responses for Control OFF and Controller A in DLC. (a) Tractor comparative results. (b) Trailer comparative results.

Both sine steer and DLC maneuvers are performed with and without the controller for comparison.

Figs. 9 and 10 give the results of sine steer maneuver. In Fig. 9, all subfigures at the left column denote the "Control OFF" maneuver, while those at the right column denote the "Controller B." The same sine steering angles are used but with a lower initial speed of 43km/h to prevent the trailer from total instability considering the limited capacity

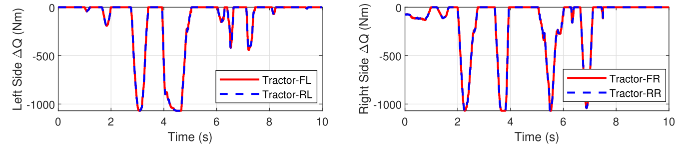

Fig. 8. Corrective torques by Controller A in DLC.

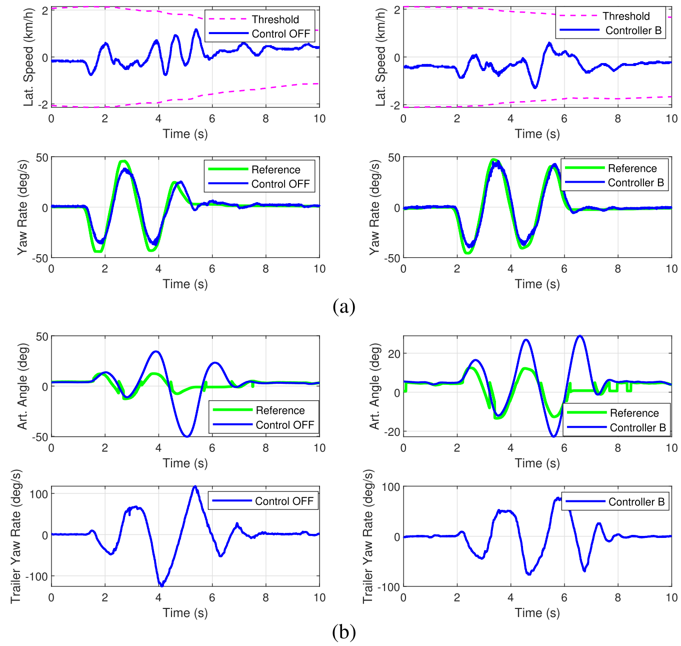

Fig. 9. Tractor-trailer responses for Control OFF and Controller B in sine steer. (a) Tractor comparative results. (b) Trailer comparative results.

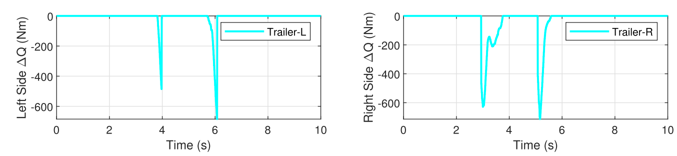

Fig. 10. Corrective torques by Controller B in sine steer.

of trailer braking. As a result, the lateral speed and yaw rate of the tractor in Fig. 9 are stable and well tracked. Fig. 9(b) compares the articulation angle and the yaw rate of the trailer. The art. angle went unstable and diverge from 0° to $-50°$ when no control engaged. However, with the active differential braking of Controller B, the art. angle is controlled within 20°, and the trailer yaw rate is reduced around 20%. The trailer corrective (braking) torques are shown in Fig. 10. It can be seen that the torque actions are less than 700 Nm and activated in short time slots, but the control is effective in sway prevention.

Figs. 11 and 12 present the results of DLC maneuver. The same DLC steering angles are used with the initial speed of 43 km/h. In the Control OFF case, both tractor and trailer showed stable and noncritical dynamics responses, but the art. angle failed to follow the reference. With the engagement of Controller B, the art. angle is controlled within 10° and followed the reference generally very well. One may note there is a constant deviation suggested in art. angle when no steering is applied (driving straight). This is from the sensor error due to the bended linkage in Fig. 3. However, the art.

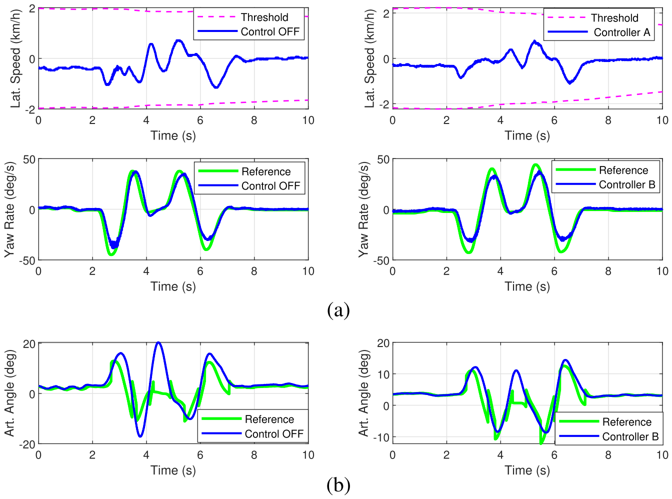

Fig. 11. Tractor-trailer responses for Control OFF and Controller B in DLC. (a) Tractor comparative results. (b) Trailer comparative results.

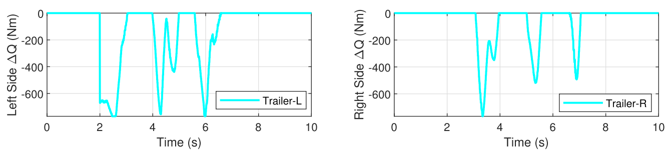

Fig. 12. Corrective torques by Controller B in DLC.

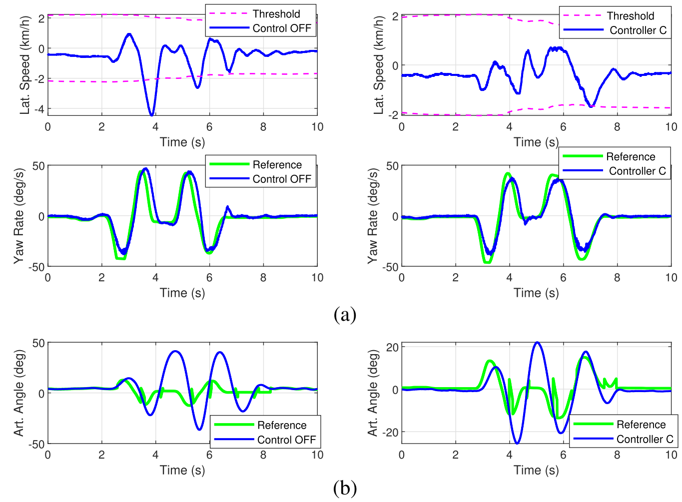

Fig. 13. Tractor-trailer responses for Control OFF and Controller C in DLC. (a) Tractor comparative results. (b) Trailer comparative results.

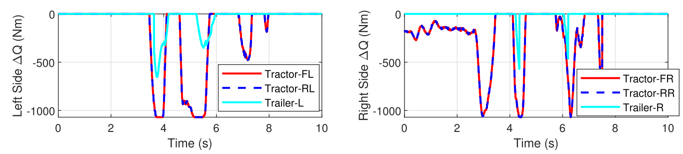

Fig. 14. Corrective torques by Controller C in DLC.

angle reference is set only when art. angle exceeds a certain threshold, which avoids undesired control. Thus, in Fig. 12, there are no unnecessary torque adjustments at the first and last 2 s.

Compared with Controller A of the same/similar maneuvers, it indicates that Controller A generally outperforms Controller B in reducing art. angle and handling the more critical situation. This is because Controller A has a much larger brake capacity than B. It uses the regenerative brake of the electric motor instead of the hydraulic brake system, which gives instant and very accurate torque feeding. In contrast, the trailer brake system can only provide limited torque and has an actuator delay.

D. Integrated Differential Braking

In this integrated differential braking strategy, six corrective torques of ΔQ_1 to ΔQ_6 are applied and the gains and bounds of the MPC and CA are listed in Table III.

Figs. 13 and 14 present the results of DLC maneuver. The Control OFF result is exactly similar to those in Fig. 7 as the same maneuvers are used for demonstration and it is discussed in Section IV-C. This helps to compare the effectiveness of Controllers A and C in the same "OFF" maneuver. First, the Controller C stabilized the tractor lateral within the bounds and prevents the trailer from sway around 20°, which is similar to the performance seen in Controller A in Fig. 7. However, looking at the details, the lateral speed, art. angle, and trailer yaw rate achieved by Controller A slightly outperformed those achieved by Controller C. Based on one single case, one cannot draw a conclusion that Controller A is superior to Controller C. There are some limitations on the trailer braking actuators since it is a drum brake. The delay issue, the limited braking torque capacity, and mismatch between the request torque and clamp force may degrade the performance of the controller or even may have side effects. From 3.5 to 4.5 s, the vehicle (tractor) is turning right, and the art. angle is swaying from positive to negative (sway from left-hand side to right-hand side). To pull the trailer back from sway and large art. angle, the controller C is applying the torque on the left side of the trailer during 3.5–4 s, as shown in Fig. 14. In addition, the tractor is also applying left side torque to prevent the tractor from oversteer. In fact, the trailer sway is not that severe during 3.5–4 s, from what can be seen from art. angle results. However, during 4– 4.5 s, trailer is about to sway to right-hand side divergently; hence, both the tractor and trailer are applying the torques on the right-hand side wheels to prevent a sway since the tractor yaw rate is not oversteer.

V. Conclusion

In this article, a reconfigurable MPC system is developed and implemented to an electrified tractor-trailer stability control. By leveraging the constrained optimization techniques, a hierarchical control scheme is formulated. The high-level MPC controller calculates corrective CG yaw moments, while the lower level CA controller is responsible for distributing the control actions to active actuators in an optimal fashion. The technique of adding/modifying constraints and weights to optimization problems makes the controller reconfigurable to different actuation systems. The proposed control system is tested on an articulated vehicle with multiple differential braking configurations. Examined with uncontrolled maneuvers for both sine steer case and double-lane change case, it shows that the controller has a very promising performance in yaw control and trailer sway prevention, as well as proven reconfigurability when moving from this configuration to another one. Moreover, given the comparison of three strategies, this article reveals a low-cost and feasible solution for articulated vehicle dynamics control in the light of the effectiveness of tractor differential braking and only one requisite sensor (art. angle) on the trailer.

The controller implementation may be further improved in future work. For instance, unwanted braking corrections at low speeds and driving in a straight line. In practice, the actuator dynamics and delay are very prominent factors that need to be taken into account when designing the stability control controller. In addition, the controller's resilience and robustness respect to the road surfaces or sensor failures are very important to real application (see [34], [35]). If the stabilizer facilities are equipped, the performance on slippery road conditions is expected to be validated.

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